

Fast flame recognition algorithm base on segmentation network

Chunyu Niu¹, Hui Guo^{1*}, Yong Wang^{2*}

1. School of Mechanical and Power Engineering, East China University of Science and Technology, Shanghai 200237, China;

2. School of Mechanical and Automotive Engineering, Qingdao University of Technology, Qingdao 266033, China

ABSTRACT

To solve the low recognition rate of the network to flame and keep the accuracy, we propose an Instance segmentation model for recognizing and locating flames more accurate This network is improved based on the deep learning model Mask R-CNN, it introduces four key components:(1) After analyzing the effects of space and channel attention, we used an efficient convolution channel attention. (2) By comparing the convolution kernel size, an optimized dilated convolution is added to the network, (3) To eliminate redundancy, reducing the depth of the backbone while guaranteeing the accuracy of the network. (4) Finally, Adding a flame extraction algorithm behind the head. Compared with Mask R-CNN, the model size is reduced by 16.3MB, and the recognition accuracy of flame is improved by 1.7%. The comparison shows that the network can also greatly improve the recognition effect of small flames.

Keywords: Flame recognition, Instance segmentation, Flame positioning, Deep Learning.

Index Terms: K.6.1 [Management of Computing and Information Systems]: Project and People Management—Life Cycle; K.7.m [The Computing Profession]: Miscellaneous—Ethics

1 INTRODUCTION

Like other disasters, fire will cause immeasurable losses once it occurs, but the difference of fire is that most fires can be avoided or reduced by manual intervention. According to the statistics of the National Fire and Rescue Administration, the number of fires in China has more than 200000 every year since 2015, resulting in economic losses of up to 4 billion yuan, and this number is still increasing year by year. Therefore, fire detection has always been a social hot-button issue.

In the past, in addition to manual detection, the visual detection methods of flame was a classification method based on SVM after image processing of flame or simple BP neural network according to the flame characteristics of color, sharp point and luminance, After comparing the detection methods based on different color spaces, or some methods based on changing color space, Ye Zihan^[1] choose to improve the original flame region constraint rules based on RGB and HSI color models to achieve accurate judgment of flame signals.

In recent years, with the development of deep learning and computer vision, many flame recognition algorithms based on convolutional neural networks have been proposed. Tong et al^[2] modified the data augmentation strategy and added the CBAM attention mechanism based on YOLOv5 network to change the loss function to improve the detection efficiency and accuracy of flame disasters, but the improved model has problems such as weak generalization ability. From the detection results, the detection

methods of neural network can be divided into image detection and instance segmentation. Among them, image detection is the most basic computer vision method, which can frame objects with rectangular frames. It has a faster detection speed, but it cannot accurately locate irregular objects; Instance segmentation is a more challenging method of computer vision. It needs to assign a pixel-level mask with a category label to each interesting instance in the image. Because image segmentation can separate the flame image, the post-processing of the flame will become more controllable, and the recognition and positioning accuracy will also be higher. Therefore, we select segmentation as the basic algorithm for flame image research.

Mask R-CNN^[3] is the most successful branch of instance segmentation. It first downsamples the binary ground truth to 28×28 grids, and then reconstructed by up sampling. And then extract the region of interest (e.g. ROIAlign) based on the frame level positioning information of the regional proposal network (RPN) or the bounding box prediction of the previous stage Finally, the extracted features are input into the head layer to get the final instance mask, and need to improve its detection speed.

The backbone used by Mask R-CNN original network is the ResNet network. ResNet network was proposed by Kaiming He in 2016. It is composed of a stem_Bloch and four stages, in which the downsample operation is carried out in the last three stages except the stage 1. but this detection network is somewhat redundant for the detection of flame recognition. Compared with resnet101 and Queryinst^[4] network, SSD and mobilenetv2^[5], it is found that the deeper network depth will not significantly improve the accuracy of detection results, compared with the increased model complexity measurement, the computational cost is very high. Obviously, the method to improve the detection accuracy in this way is not applicable. The similarity between mobilenetv3 and the ConvNext^[6] network is spindle structure used in the bottleneck structure, ResNet is first reduced to the original $1/4$ and then enlarged, while the spindle bottleneck is first enlarged and then reduced in different proportions. This method can increase the network width inside the block. However, the effect in the application of flame recognition is not ideal.

In this paper, we propose a lightweight feature extraction network to replace the traditional feature extraction network. Firstly, prune the basic structure of the network to reduce the network depth on the premise that the extraction effect is basically unchanged. Secondly, the dilated convolution and attention mechanism network are introduced to improve the ability to learn and focus on useful channels of the network, and change the activation function strengthen the effective segmentation ability of the flame region in the image. Finally, a flame ignition point identification network is added to the detection mask at the end of the network to achieve more accurate identification and positioning of the flame.

2 THEORY

The visual detection model based on deep learning can be divided into two parts: feature extraction network and feature detection network. To improve the accuracy of flame recognition and reduce the complexity of the network, a lighter feature extraction network and a two-level cascade detection network are proposed for the

¹ 2369352598@qq.com

^{1*} huiguo@ecust.edu.cn

^{2*} ngecut@163.com

shape and characteristics of flame. The feature extraction network structure is shown in Figure 1.

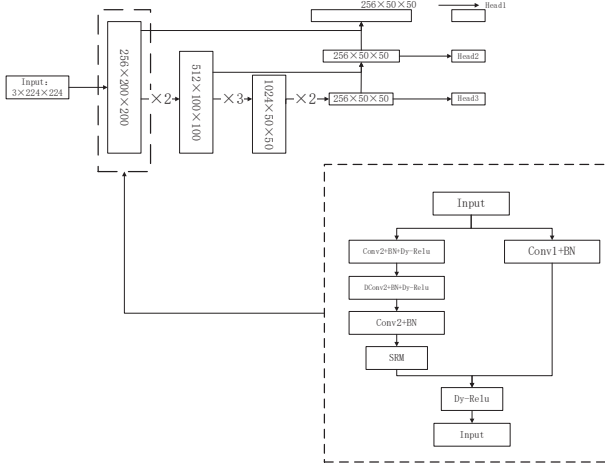


Figure 1: Network structure

Compared with ResNet network, we use three feature convolution layers to extract flame, it has smaller network depth and faster operation speed.

2.1 Basic network structure

Because of the internal characteristics of the flame are not obvious, and when we find that increase the network depth but not significantly improving the detection accuracy, we try to reduce the network depth, it is found that the detection accuracy will not change greatly, and the complexity of the network can be reduced. Keeping the basic structure of ResNet network, we make a large number of experiments to reduce the number of network layers. At last, the network is changed from the original four-layer network to three-layer network, and the three-layer characteristic network is changed to 2,3,2, which ensures the accuracy, also reduces the complexity of the network to a certain extent.

2.2 Network Attention Selection

Because the flame background is more complex and disturbing, attention mechanism is used to make the model better aware of the future flame, and the flame characteristics are sensitive to the change of channel, so channel attention mechanism and spatial attention mechanism are used to test whether they can improve the network respectively.

Channel attention was first raised in SENet^[7]. Its core is the squeeze-and-excitation (SE) module, which collects global information, captures channel relationships, and improves expressive ability.

To avoid the model being too complex, Lee et al.^[8] proposed the lightweight style-based recalibration module (SRM). SRM combines style transfer with an attention mechanism. Its main contribution is style pooling which utilizes both mean and standard deviation of the input features to improve its capability to capture global information. It also adopts a lightweight channel-wise fully-connected (CFC) layer, in place of the original fully-connected layer, to reduce the computational requirements.

Woo et al. proposed the CBAM^[9] attention module (CBAM, Convolutional Block Attention Module), which, unlike the two above, combines channel attention with spatial attention. and by introducing global pooling to utilize the spatial global information, CBAM attention can be performed on both channel and spatial dimensions.

After comparison, it is found that due to the spatial characteristics of the flame is not obvious, the recognition accuracy is reduced to a certain extent when using spatial attention, and the recognition accuracy is improved when use SRM channel attention alone.

2.3 Dilated convolution

Full convolutional networks (FCN) for image segmentation mainly has two key parts. One is to reduce the image size and then increase the receptive field by convolution and pooling, and the other is to expand the image size by up sampling. In the process of first reducing and then increasing the size, there will be information loss. By keeping the size of the convolution core unchanged and increasing the coverage of the convolution core, the dilated convolution^[10] can ensure that the receptive field is increased without increasing the amount of calculation, and more detailed information is retained compared with the up sampling. The calculation formula of equivalent core size is as follows

$$K = (d - 1)(k - 1) + k \quad (1)$$

Where d is the dilated rate, k is the real size of dilated convolution, and K is the equivalent core size. However, when using dilated convolution, not all the pixels covered by it are used for calculation, so using dilated convolution kernel to improve the receptive field will lose the continuity of information. This method of increasing receptive field has a great impact on most pixel level dense prediction tasks, but for flame segmentation, the characteristics of flame are not the main distinguishing features in terms of its continuity, and most objects are similar to the fire in color and shape have different continuity, Special regularity (periodicity, etc.) and more recognizable similarity than flame. Therefore, this paper uses dilated convolution in the feature extraction network and puts it into each block of the ResNet-light. The results show that the backbone using dilated convolution has a significant effect on improving the ability of flame feature extraction, after adjusting the direction and size of dilated convolution according to the characteristics of flame, it is found that the most suitable low layer convolution is 7×5 , and dilated convolution is not used in the third layer network to retain more deep information. The final network structure diagram is shown in Figure 1, in which dconv2 is the convolution layer using dilated convolution, and the downsampling layer (conv + BN on the right) is only used in the first layer of each block.

2.4 Activate function selection

Because the results of using Relu activation functions in neural networks are usually better than those of using other activation functions, the main activation functions used in neural networks are Relu activation functions and variants based on Relu activation functions, such as LeakyRelu, PRelu, etc. However, these activation functions are static activation functions, that is, their function forms are fixed in any case, Yinpen Chen et al. proposed a dynamic activation function, Dynamic-Relu^[11], based on Relu activation function, which shape can change during training.

$$y_c = f_{\theta(x)}(x_c) = \max_{1 \leq k \leq K} \{a_c^k(x) x_c + b_c^k(x)\} \quad (2)$$

As shown in formula (2), a and b are two variable parameters respectively. When $x > 0$, $a=1$, and $b=0$, the function is Relu. The Dynamic-Relu activation function is equivalent to changing the values of a and b within a certain range according to the difference of input data during training, which makes the expression ability of the activation function stronger, and is applicable to lightweight networks. The experiment shows that the detection accuracy of the network is greatly improved after using the Dynamic-Relu activation function to replace Relu activation function.

2.5 Fire point identification

For flame identification, its main purpose is to extinguish the fire, so the identification of flame ignition point is particularly important. The common method is to use the flame centroid as the output of the ignition position, but this method is not accurate and not universal. To ensure the identification accuracy and accurately locate the ignition point position, we also add the position identification module of the flame ignition point on the basis of the model. Through analyzing the principle of flame formation, As well as the summary of the position of the flame ignition point under different flame sizes, it is finally found that the upper part of the flame usually interferes with the judgment of the flame ignition point, and it is also more inaccurate to use the centroid as the ignition point under a larger flame area. Therefore, a recognition algorithm for the ignition point is designed, As shown in formulas 3 and 4, which sets the weights for the ratio k of the upper and lower parts of the flame, the areas of the flame, and the ratio l of the flame length to width, respectively, ω is the corresponding weight, based on the centroid calculation formula, the ignition point of the flame is displayed within the mask range, and output according to the size of the flame.

$$X = \frac{\sum_i m_i x_i \omega_1 l k}{\sum_i m_i} \quad (3)$$

$$Y = \frac{\sum_i m_i y_i \omega_2 l k}{\sum_i m_i} \quad (4)$$

3 DISCUSSION

Flame data comes from open source project “<https://github.com/steffensbola/furg-fire-dataset>” Through the frame sampling of those videos, a total of 700 frames containing more than 10 kinds of flame scenes are obtained, as shown in Figure 2, the segmentation and annotation are completed with labelme.



Figure 2: Partial sample dataset

3.1 Optimizer selection

Compared with SGD, AdamW^[12] has faster convergence speed. We set the total learning rate to 0.001, adjust the learning rate of backbone part to 0.1 times of the total learning rate, and increase weight decay base on Adam to make it have better convergence..

Finally, the 700 images were trained 4200 times, and make a comparison by training the multi model.

3.2 Compare the effects of different algorithms

To explore the performance of the model in this paper, we compare the network model ResNet-Fire proposed in this paper with Mobilenet 、 SSD-VGG 、 Solov2^[13] 、 Queryinst 、 Convnext network and the backbone based on ResNet. As shown

in Table 1, the average accuracy is mask AP, FPS is measured based on RTX2080Ti GPU, and the batch size is set to 1.

Table 1: Comparison with different methods

Method	Weight/M B	MAP/%	speed/FPS
MobileNet	22.7	56.9	31.5
SSD-VGG	35.2	65.2	22.9
Solov2		64.5	12.1
Queryinst	172.2	66.3	6.9
Convnext	47.7	65.7	18.1
ResNet	43.8	69.1	21.1
ResNet-fire	27.5	70.8	23.4

The results of model comparison experiments show that when MobilenetV2 is used as the backbone network, its network occupies very low memory, and its detection speed is fast, but its accuracy rate is far lower than that of Mask-R-CNN model based on other backbone networks, which can not meet the accuracy requirements of flame recognition. When SSD-VGG network is used as the network backbone, although the calculation amount is reduced compared with Resnet system, the accuracy also decreases in a ladder way, When the ResNet network is used as the backbone model, the accuracy will be greatly improved, but the memory consumption will also increase.

Model comparison experiment show that compared with other networks, the ResNet-fire backbone network proposed in this paper has improved the detection accuracy of the network by nearly 10 percentage points when compared with Mobilenet and SSD-VGG16 as the network backbone by adjusting the ResNet network layer structure, increasing the hole convolution, introducing the SRM attention mechanism and using the Dynamic-Relu activation function. Finally, not only the model weight is reduced by 16.3MB, but also the detection accuracy is improved by 1.7% compared with the traditional network.

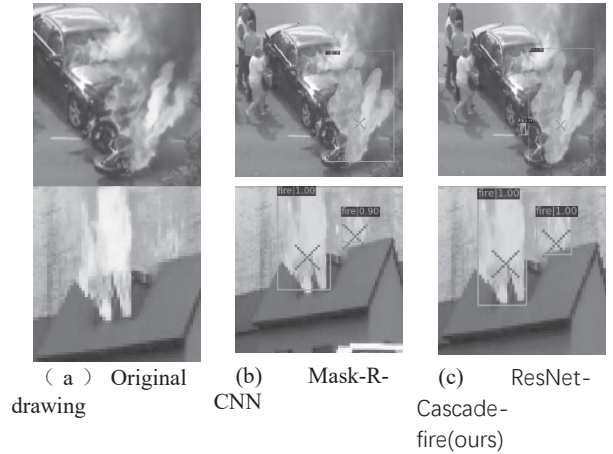


Figure 3: Comparison of model detection capability

As shown in Figure 3, compared with figure (b) Mask R-CNN network model, (c) the model proposed in this paper can detect the smaller flame in the lower left corner of the wheel and more accurately to the weak flame on the right side of the house roof, which proves that with the improvement of detection accuracy the method proposed in this paper also has higher sensitivity to the small flame in complex background and the flame with unclear characteristics.

4 CONCLUSION

In the field of fire detection, how to improve the detection speed and the positioning accuracy of fire is the primary task. The existing basic network has no research related to flame recognition. Therefore, applying the existing network in practical application will make the network too complex, and the final detection effect is often poor. In this paper, we design a lightweight network model that can effectively identify the flame based on the characteristics of the flame, compared with the traditional network model, we change basic structure of the network to reduce the network depth on the premise that the extraction effect is basically unchanged. introduce the dilated convolution and attention mechanism network to improve the ability to learn and focus on useful channels, and change the activation function strengthen the effective segmentation ability.

The network proposed in this paper improves the detection accuracy by 1.7% and the network weight is reduced by 16.3MB. In addition, based on the flame shape, we design an algorithm which can accurately identify the fire point of the flame and add it to the network, the algorithm does not require any pre-processing or pre-training, and almost no harm to the running speed. Our method provides a more effective idea for the identification and positioning of flame, and it also has great significance to the fire-fighting work.

The improved network model has smaller memory and faster detection speed. However, for flame detection, the detection accuracy is very important, in the future work, we will improve the detection accuracy based on the algorithm in this paper to meet the accuracy requirements of flame detection.

REFERENCES

- [1] Ye Zihan. Dynamic flame recognition algorithm based on multi-feature fusion (in Chinese). *Information Recording Materials*, 23(06):199-202. December 2022. doi: 10.16009/j.cnki.cn13-1295/tq.2022.06.059
- [2] T Yang, S Xu, W Li, H Wang, G Shen and Q Wang. A smoke and flame detection method using an improved yolov5 algorithm. (Guiyang, China 17-22 July 2022), 2022 IEEE International Conference on Real-time Computing and Robotics (RCAR), pages 366-371, IEEE, July 2022. doi: 10.1109/RCAR54675.2022.9872297
- [3] He K, Gkioxari G, Dollár P, Girshick R. Mask R-CNN. (Venice, Italy, 22-29 October 2017), 2017 IEEE International Conference on Computer Vision (ICCV), pages 2980-2988, IEEE January 2017. doi: 10.1109/ICCV.2017.322
- [4] Yuxin Fang, Shusheng Yang, Xinggang Wang, Yu Li, Chen Fang, Ying Shan, Bin Feng, Wenyu Liu. Instances as queries. (Montreal, QC, Canada 10-17 October 2021), IEEE/CVF International Conference on Computer Vision (ICCV), pages 6890-6899, IEEE February 2022. doi: 10.1109/ICCV48922.2021.00683
- [5] Howard A, Sandler M, Chen B, Wang W, Chen L C, Tan M, Chu G, Vasudevan V, Zhu Y, Pang R, Adam H, Le Q. Searching for MobileNetV3. (Seoul, Korea (South) 27 October 2019 - 02 November), 2019 IEEE/CVF International Conference on Computer Vision (ICCV), pages 1314-1324, IEEE February 2020. doi: 10.1109/ICCV.2019.00140
- [6] Z Liu, H Mao, C-Y Wu, C Feichtenhofer, T Darrell and S Xie. A. ConvNet for the 2020s. (New Orleans, LA, USA 18-24 June 2022), 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 11966-11976, IEEE September 2022. doi: 10.1109/CVPR52688.2022.01167
- [7] Hu J, Shen L, Sun G. Squeeze-and-Excitation networks. (Salt Lake City, UT, USA 18-23 June 2018), 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 7132-7141, IEEE December 2018. doi: 10.1109/TPAMI.2019.2913372
- [8] Hyunjae Lee, Hyo-Eun Kim et al. SRM: A Style-Based Recalibration Module for Convolutional Neural Networks. (Seoul, Korea (South) 27 October 2019 - 02 November), 2019 IEEE/CVF International Conference on Computer Vision (ICCV), pages 1854-1862, IEEE February 2020. doi: 10.1109/ICCV.2019.00194
- [9] Woo SH, Park J, Lee JY, Kweon IS. CBAM: Convolutional block attention module. In *Lecture Notes in Computer Science*, volume 11211, pages 3-19. 2018
- [10] Chen, LCE, Zhu, YK, Adam, H. Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation. In *Lecture Notes in Computer Science*, volume 11211, pages 833-851. 2018.
- [11] Jiong Si, Sarah L. Harris et al. A Dynamic ReLU on Neural Network. (Dallas, TX, USA 12-12 November 2018), 2018 IEEE 13th Dallas Circuits and Systems Conference (DCAS), pages 1-6, IEEE, January 2019. doi: 10.1109/DCAS.2018.8620116
- [12] Ilya Loshchilov, Frank Hutter. Decoupled weight decay regularization [EB/OL]. [2019-1-4]. <https://arxiv.org/abs/1711.05101>
- [13] Xinlong Wang, Rufeng Zhang et al. SOLO: A Simple Framework for Instance Segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44:8587-8601, September 2021. doi: 10.1109/TPAMI.2021.3111116